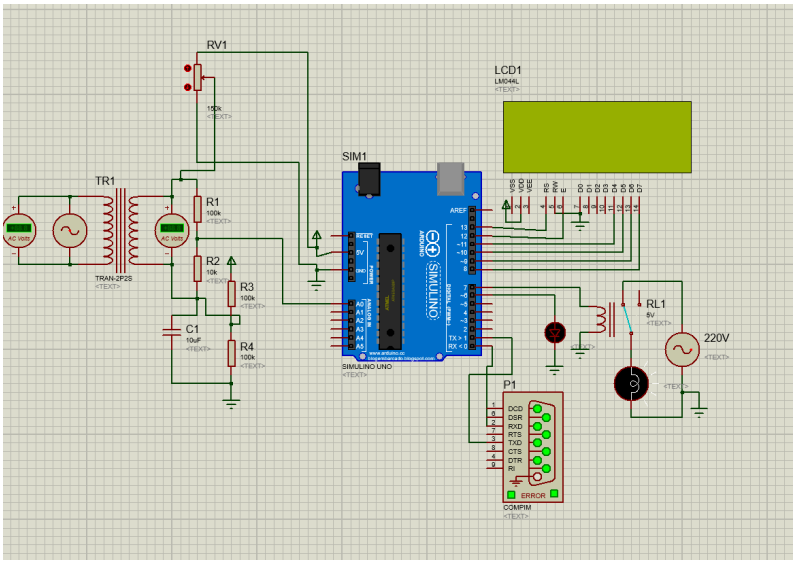
Report: Over & Under Voltage Protection System (MD. Atair Rahman Alvi)



Design approach: The Arduino cannot take mains directly,y and its ADC only reads 0 to 5 V. So the signal goes through a transformer (isolation and step-down), a 100k/10k divider, and a bias network (R3, R4, C1) that centres the AC wave at 2.5 V. The EmonLib library calculates the RMS voltage from pin A0. I tuned the calibration constant (187) until the LCD matched the reference voltmeter. The logic is simple: 198 V to 242 V (The safest voltage range for standard 220V electronics is typically between 198V and 242V) is normal, al and the relay (D7) stays on. Outside that range,nge the relay switches off and the LED turns on. After a fault, the relay only comes back once the voltage has stayed between 192 V and 248 V for 5 seconds, so it does not chatter near the limits. The relay is off at start-up.

Test	Expected
Normal 220 V	Relay ON, LCD shows NORMAL VOLTAGE
Above 242 V	Relay OFF, LED ON, LCD shows OVER VOLTAGE
Below 192 V	Relay OFF, LED ON, LCD shows UNDER VOLTAGE

What is not done, and why: I couldn't establish the computer connection because the Com0Com software, which is used to create virtual COM ports for serial communication at 9600 baud, was not working properly on Windows 11 due to a driver issue. As a result, I couldn't complete the Virtual Terminal communication part of the task.